

Clinical Risk Assessment Report: Patient 49

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 42.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.93x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.36 [Bottom 9%] (+5.8%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.58 [Avg (73th %ile)] (+3.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.02 [Top 13%] (+3.1%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.74 [Avg (72th %ile)] (+2.7%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 1.15 [Top 4%] (+2.4%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.4 [Bottom 2%] (+1.9%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.01 [Top 15%] (+1.9%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 11%] (+1.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

